

Feral Pig / Pest-Control Patent & Business Buildout

Project roadmap, workstreams, gates, and current risk posture

Working document only. Patentability, inventorship, filing strategy, regulatory status, and legal conclusions require appropriate licensed counsel and primary-source verification.

Current posture

- **Set 1 only is active.** Target architecture remains 20 claims: 3 independent + 17 dependent. Set 2 is locked until Set 1 is complete and a distinct-invention justification is written.
- **Provisional filing is gated by disclosure quality.** The specification must support the intended later claim scope with written description and enablement; finished claims are not the filing-date gate.
- **Patentability risk is now material.** Initial triage identified earlier image-recognition/euthanasia, audio+video feral-swine recognition/toxicant activation, and species-specific wild-boar poison-feeder disclosures.
- **No external technical disclosure without NDA or filing.** Maintain a disclosure log and settle inventorship/conception records before equity or IP assignment decisions.

Workstream and gate structure

1. Evidence & source ledger

Ingest every technical/business/legal source; classify primary vs. secondary; record provenance, date, relevance, and follow-up. Treat the comedy transcript as a lead index, not proof.

2. Enabling technical disclosure

Draft architecture, sensor modalities, recognition pipeline, group/member identification logic, confidence thresholds, failure modes, selective actuation/delivery, lethal modalities, safety interlocks, logging, hardware/software, deployment variants, and examples.

3. Set 1 claim architecture

Maintain the claim inventory as single source of truth. Draft the mid-scope independent only after prior-art charting identifies defensible technical distinctions; then draft the 17 dependents.

4. Patentability search

Run element-by-element claim charts against known references, then structured USPTO Patent Public Search / Google Patents / scholarly and product literature searching. Distinguish patentability from freedom to operate.

5. Filing package

Specification, drawings, provisional cover information, inventorship check, disclosure log, source-format validation, and licensed patent-attorney review before submission.

6. Regulatory & deployment

Map EPA/pesticide, wildlife, state/local, operator, carcass/environmental, and site-specific constraints. Patent scope and lawful field deployment are separate questions.

7. Business model

Segment government agencies, agriculture/landowners, wildlife-control operators, infrastructure sites, OEMs, and channel partners. Test hardware sale, managed service, licensing/OEM, data/monitoring, and consumables/service revenue models.

8. Funding/company/IP

Conception/inventorship record; entity and IP-assignment plan; federal/state grant pipeline; budget gates for each nonprovisional or restriction-driven path.

9. Experimental partnerships

Before filing: target identification and NDA-ready outreach only. After filing/NDA: pilot scope, regulatory pathway, safety protocol, data rights, background/foreground IP, publication review, and success metrics.

Immediate technical disclosure questions

- How is a local population/sounder defined and bounded in space and time?
- How are known individuals enrolled, re-identified, added, removed, or marked unknown?
- What evidence is sufficient to conclude that all required members are present, and what confidence/coverage threshold is used?
- What happens when visibility/audio coverage is incomplete, classifications conflict, a non-target appears, or a new/unknown pig appears?
- Exactly what event gates lethal actuation or access to a generally toxic agent, and how is the system made fail-safe?
- Which concrete lethal modalities are enabled by the disclosure, rather than merely named by function?
- What records are logged so the system can be audited after an actuation decision?

Initial patentability triage - priority references

[US10368539B2 - Species specific extermination device](#)

Image/reference comparison is used to decide whether to energize electrodes and euthanize a target animal; this is close to the generic Claim 1 architecture.

[US20210315186A1 - Intelligent dual sensory species-specific recognition trigger system](#)

USDA publication expressly addresses feral swine, audio+visual recognition, autonomous triggering, estimated quantity, and wildlife devices that can control toxicant delivery.

[US20140261201A1 / US9295225B2 - Species specific feeder](#)

Species-specific wild-boar feeder disclosure includes population-control compositions, poison, and short-term elimination of a local population.

Business-model hypothesis to test

The strongest commercial position may be a recognition-and-decision control layer that can gate multiple approved control mechanisms, rather than a poison-specific product. That hypothesis must be tested against prior art, customer procurement, regulatory burden, liability, efficacy, serviceability, and the ability to produce auditable low-false-positive decisions.